Multi-Criteria Decision Analysis for Strategic Supplier Selection: A Comparative Study of AHP, ANP, and TOPSIS in the Electronics Industry*

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Abstract

Electronics industry is a complex industry, and the system of supplier selection is linked with multiple contradictory conditions, qualitative selection and uncertainty. Even the more conservative practices are not even inclined towards these complications. In this work we use three Multi-Criteria Decision-Making (MCDM) methods (AHP, ANP, and TOPSIS) to classify the suppliers in a structured manner based on some criteria, such as Cost, Quality, Delivery, Technology, and Financial Stability. The comparative analysis indicates the strong points, the weaknesses and usability of the two approaches and is a good structure in the well-crafted and clear strategic sourcing.

1 Introduction

The capacity to choose a strategic supplier is one of the important aspects of organizational performance, sustainability, and competitive advantage in the contemporary and dynamic global supply chain environment. This is better understood in the electronics industry whereby there is a complex supply chain, a massive product recall, and a rising e-waste issue. The choice of suppliers is efficient to allow complying with the regulations, minimize operational risks, enhance

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customer satisfaction, and advance the concept of the circular economy by the opportunity to retrieve the value of the products returned [5, 10]. However, the decision-making process of choosing the suppliers is complicated by nature, and one has to consider different and sometimes conflicting criteria, based on uncertainty and the lack of information. Most of the traditional methods of selection do not consider these interdependencies, and this will result in the making of decisions that may not be optimal. Other issues complicate the decision-making process, and these include financial constraints, technicalities, volatility in the market, and lack of skilled labor. On this basis, it is with a sense of urgency that systemic, methodological, sound frameworks are needed to bring both quantitative and qualitative variables into one [24, 18, 16].

Today, the choice of suppliers is increasingly made by leveraging digital procurement platforms, analytics of data, and intelligent decision-support systems. The use of MCDM methods such as AHP, ANP, and TOPSIS in the assessment of suppliers can enhance transparency, consistency, and strategic decision-making in an electronic supply-chain ecosystem. This study thus not only provides an extension to existing supplier-selection research, but also contributes to the development of intelligent E-Business Information Systems aimed at supporting digital transformation and sustainable procurement practices.

2 Multi-Criteria Decision-Making (MCDM)

Multi-Criteria Decision-Making (MCDM) methods provide structured approaches that are used to assess alternatives based on more than one criterion, where alternative criteria may be in conflict [19, 22]. In supplier selection, where financial stability, technological capability, delivery reliability, quality, and cost are all qualitative and quantitative criteria that need to be taken into account, MCDM techniques provide a systematic basis for integrating these factors in a single, transparent evaluation process [4, 9]. Each of the three methods used in this study deals with this challenge differently: AHP through pairwise comparison in a hierarchy, ANP through interdependency modelling in a network, and TOPSIS through geometric distance from ideal solutions.

3 Literature Review

The process of strategic selection of suppliers is a conclusive performance determinant of organizational business in the modern global supply chain management (SCM) landscape. This is particularly applicable in the electronics industry, which is typified by complex supply chains, high product recall, and increasing e-waste problems [5, 11, 8]. The selection of good suppliers enhances compliance with environmental requirements, minimizes operational risk, improves customer satisfaction, and is likely to reclaim the value on any returned product, hence supporting the objectives of the circular economy.

3.1 Problems in Supplier Selection

The procedure is complicated because there are many, often interdependent criteria, unclear or incomplete information, and the qualitative character of many assessment factors. These interdependencies are often ignored by historical selection practices, resulting in suboptimal decisions [13, 12]. In addition, systematically acceptable methodological designs are required since most studies rely on demonstrative numerical illustrations. Other issues such as financial constraints, regulatory conditions, technical conditions, unstable markets, and unavailability

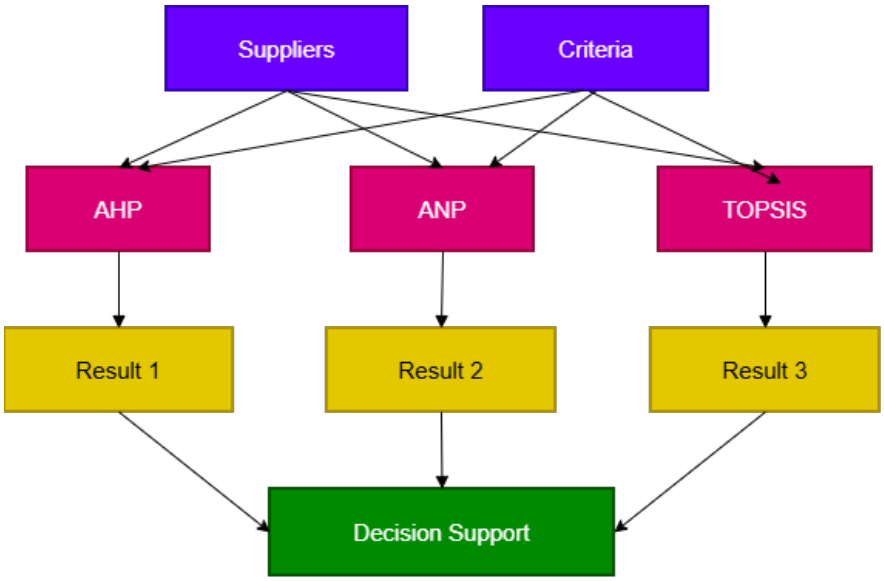


Figure 1: **Framework for Supplier Evaluation Using MCDM Techniques**

of skilled workforce in other markets such as Taiwan also act as aggravating factors in the management of suppliers.

3.2 MCDM Approaches in Supplier Selection

MCDM techniques are widely adopted across supply chain management, manufacturing, and procurement due to their capability to combine quantitative metrics with qualitative expert judgment [1, 7]. Among these, AHP remains the most frequently applied method, valued for its consistency-checking mechanism and intuitive hierarchical structure [17, 2]. However, its assumption of criteria independence limits its applicability in complex industrial contexts where criteria interact.

To address this, researchers have explored hybrid and extended approaches. ANP relaxes the independence assumption by modelling feedback and interdependencies between criteria through supermatrix construction, making it particularly suited to electronics supply chains where quality, technology, and delivery are interrelated [4, 6]. TOPSIS complements both methods by providing an efficient geometric ranking mechanism that is commonly integrated with AHP- or ANP-derived weights to produce final supplier rankings [21].

Recent studies have demonstrated the value of combining these methods. Hesami [4] applied a hybrid AHP-TOPSIS approach under rough uncertainty in electronics supplier selection, while Menon and Ravi [13] showed that AHP-TOPSIS integration improves transparency and consistency in sustainable supplier evaluation. Despite this body of work, direct three-way parallel comparisons of AHP, ANP, and TOPSIS on an identical case with full numerical traceability remain underrepresented, particularly within the electronics industry context. This paper addresses that gap.

3.3 Key MCDM Methodologies

1) Analytic Hierarchy Process (AHP): AHP was developed by Thomas L. Saaty and classifies criteria into a hierarchy, then makes pairwise comparisons to derive weights [17]. Its consistency tests reduce subjectivity, but it assumes independence of factors and uses precise numerical inputs, which may fail to capture interdependencies.

2) Technique for Order Preference by Similarity to Ideal Solution (TOPSIS): TOPSIS ranks alternatives by how close they are to a best solution and how far from a negative-ideal solution [21]. It is valued for conceptual clarity and computational efficiency and is commonly used together with AHP, where AHP provides criterion weights and TOPSIS produces final rankings.

3) Fuzzy Set Extensions: Fuzzy logic can overcome linguistic vagueness and subjectivity. Fuzzy AHP accepts qualitative criteria, Fuzzy TOPSIS measures alternatives with fuzzy numbers, and Fuzzy ANP considers interdependencies and feedback between criteria [25]. Interval Type-2 Fuzzy Sets provide better treatment of uncertainty and have found use in algorithms such as IT2FAHP, IT2FTOPSIS, and IT2FVIKOR.

4) Hybrid Approaches: A combination of MCDM methods capitalises on the strengths of each method while reducing its weaknesses. Examples include ANP-TOPSIS under rough uncertainty, fuzzy AHP-fuzzy VIKOR, and AHP-Entropy with fuzzy TOPSIS-ELECTRE [4].

3.4 Supplier Selection in the Electronics Industry

The electronics industry, with its complicated supply chain, high rate of returns, and rapid technological change, requires sophisticated supplier selection [14]. Key evaluations include product quality, delivery, cost, technology, design, capacity, experience, reputation, responsiveness, service, ICT, and environmental and social responsibility. Green and resilient supplier selection is becoming more important, especially in Vietnam, India, Taiwan, and China, to counter disruptions and meet sustainability objectives [3].

3.5 Supplier-Selection Criteria

The criteria for supplier evaluation fall into several dimensions:

Traditional Criteria: Cost, quality of materials, delivery reliability, lead times, and financial stability; sub-factors include defect rates, consistency, delays, and proactive pricing information.

Green/Environmental Criteria: Environmental management systems, green technology, hazardous material handling, energy efficiency, recycling, green packaging, eco-design, pollution control, and resource optimisation [15].

Social Criteria: Occupational safety, human rights, corporate social responsibility, and workforce training.

Strategic/Alliance Criteria: Knowledge management, risk sharing, cultural fit, and IT integration.

Ethical Criteria: Code of conduct, transparency, and conflict-of-interest management, regarded as part of sustainable supply chains.

Resilience Capabilities: Absorptive, adaptive, and restorative capabilities; back-up suppliers; secure locations; and information continuity provisions [23].

3.6 Future Directions

New studies should incorporate Artificial Intelligence (AI) and machine learning with MCDM models to facilitate real-time, data-driven supplier analysis. Industrial and geographical expansion will improve generalisability, and more sophisticated fuzzy set approaches (e.g., neutrosophic, hesitant fuzzy, Interval Type-2) can better deal with uncertainty. Simplification of complex models, focus on social performance measures, creation of user-friendly software, and long-term cost-benefit analysis are vital to scalable sustainable supplier selection in the electronics industry.

3.7 Research Gap and Contribution

Though prior studies have integrated AHP, TOPSIS, fuzzy MCDM, and hybrid approaches in evaluating suppliers, many are dedicated either to hybrid optimisation or uncertainty modelling. Few studies exist that allow a clear comparison of AHP, ANP, and TOPSIS on the same supplier-selection scenario in the electronics industry with full numerical traceability.

To fill these gaps, this paper proposes: (1) a transparent comparative implementation of AHP, ANP, and TOPSIS in the same decision environment; (2) a structured evaluation framework tailored to the electronics industry; (3) an illustration of both independence-based and network-based MCDM approaches; and (4) a discussion of how these methods contribute to data-driven decision-making within E-Business Information Systems.

4 Methodology

The combination of the Analytic Hierarchy Process (AHP), the Analytic Network Process (ANP), and the Technique of Order Preference by Similarity to Ideal Solution (TOPSIS) can be used to build a sound framework in the decisionmaking techniques to be applied to complex decision-making related to suppliers selection. These techniques which are commonly used in hybrid forms allow structured analysis of various opposing criteria, allow information which is uncertain or inaccurate, and include the subjective aspect of qualitative analysis.

4.1 Methodological Framework

In this study, AHP, ANP, and TOPSIS are applied concurrently to the same supplier selection problem, enabling direct comparison of their outputs under identical input conditions [25]. AHP provides the primary criteria weight vector through hierarchical pairwise comparison; ANP extends this by modelling inter-criteria dependencies via supermatrix construction; and TOPSIS uses the resulting weights to rank suppliers by geometric proximity to ideal solutions [1]. The three methods form a complementary evaluation chain in which each method's output can be cross-validated against the others, strengthening the robustness of the final recommendation.

4.2 Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) was created by Thomas L. Saaty; this approach is a hierarchical method of solving multi-criteria decisions using both quantitative data and qualitative judgments [26]. AHP systematically incorporates human judgment into decision-making by breaking down a complex decision problem into a hierarchy of objectives, criteria, sub-criteria, and alternatives.

The AHP methodology proceeds as follows:

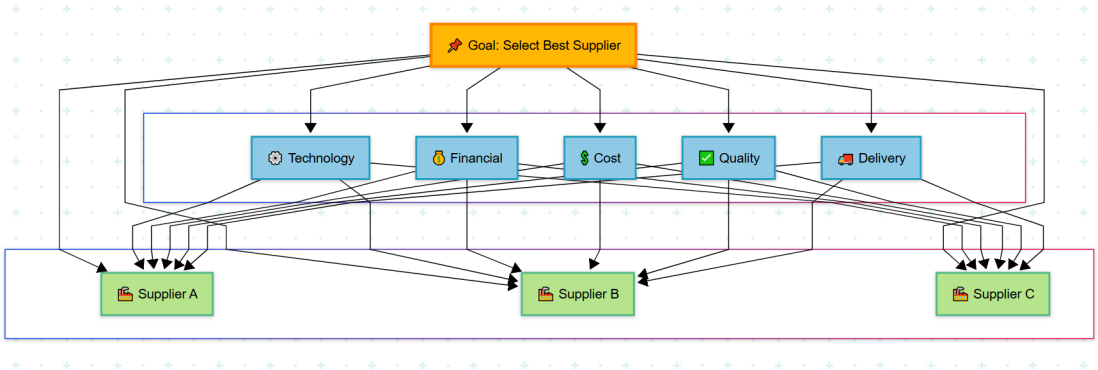


Figure 2: **Analytic Hierarchy Process Diagram**

1) Problem Definition and Objective Formulation: Clearly state the decision-making objective and scope.

2) Hierarchy Construction: Build a hierarchy with the overall objective at the top, main criteria and sub-criteria in the middle, and alternatives at the bottom.

AHP offers quantitative rigour to qualitative judgments and makes complex decisions tractable. Limitations include the assumption that criteria are independent, which may not hold in complex real-life situations [21].

4.3 Analytic Network Process (ANP)

The Analytic Network Process (ANP) is an extension of AHP that considers dependency and feedback among decision criteria [4]. Also introduced by Saaty, it describes decision problems as networks of interrelated clusters and elements, making it best applicable to complex, real-world decision situations.

ANP methodology entails:

1) Objective and Network Formulation: Determine the decision goal, criteria, and sub-criteria, and arrange them in inter-related groups to show dependencies.

2) Pairwise Comparison Collection: Gather expert opinions of relative importance among and within clusters.

3) Fuzzy Conversion (Optional): When using Fuzzy ANP, linguistic judgments can be converted to triangular fuzzy numbers to represent uncertainty and vagueness.

4) Weight Calculation: Normalise the pairwise comparisons to ensure weights sum to unity.

5) Supermatrix Construction: Construct the supermatrix, storing all interactions between elements, with each block indicating the effect of elements in one cluster on those in another [4].

6) Limit Supermatrix Computation: Raise the supermatrix to repeated powers until convergence, to obtain stable weights representing all interdependencies.

7) Priority Vector Derivation: Calculate final weights of criteria and alternatives, and perform a consistency check similar to AHP.

Unlike AHP's strict hierarchy, ANP models a network of interdependent clusters. Orange (bold) colored arrows show inner-cluster dependencies: Technology influences Quality (weight 0.40) and Financial Stability influences Delivery (weight 0.30), encoded in the unweighted

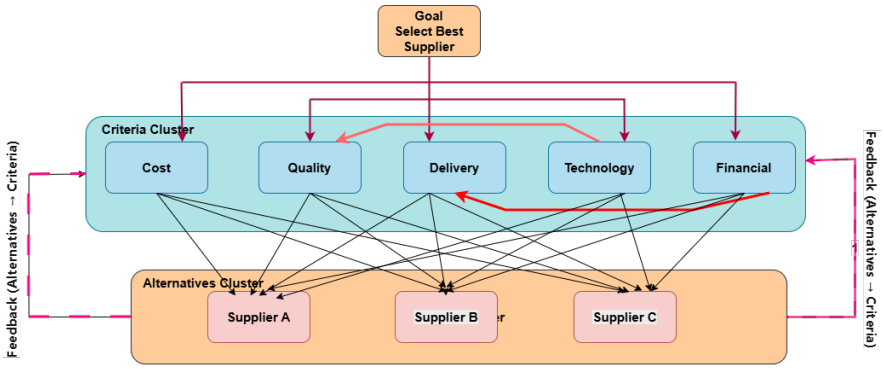


Figure 3: **Analytic Network Process Diagram**

supermatrix. Other dashed arrows show feedback loops from the Alternatives cluster back to the Criteria cluster — the defining structural feature distinguishing ANP from AHP.

ANP represents a more realistic and complete model because it includes interdependencies and feedback loops. It can further be augmented with fuzzy theory (FANP) to increase its ability to characterise uncertainty and support group consensus.

4.4 Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

TOPSIS is a distance-based MCDM approach, founded on the premise that an optimal alternative is closest to the positive ideal solution (PIS) and farthest from the negative ideal solution (NIS) [21]. Its geometrical nature, computational performance, and absence of complexities make it widely adopted in real-world decision-making.

TOPSIS procedure includes:

1) Decision Matrix Construction: Define a matrix with alternatives as rows and criteria as columns.

2) Normalisation: Convert diverse units into a dimensionless form to enable equitable comparison.

3) Weighted Normalisation: Multiply the normalised matrix by criterion weights (obtained from AHP/ANP).

4) Identification of PIS and NIS: Determine the best and worst values for each criterion across alternatives. For benefit criteria, PIS = max and NIS = min; for cost criteria, PIS = min and NIS = max.

5) Separation Measure Computation: Calculate the Euclidean distance of each alternative from the PIS and NIS.

6) Closeness Coefficient (CC_i):

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{1}$$

where d_i^+ and d_i^- denote distances from PIS and NIS, respectively [14, 20].

7) Ranking: Rank alternatives by descending CC_i; higher values indicate superior performance.

Fuzzy TOPSIS and Rough TOPSIS extensions can model linguistic uncertainty and data imprecision, further increasing their utility in complex, uncertain environments.

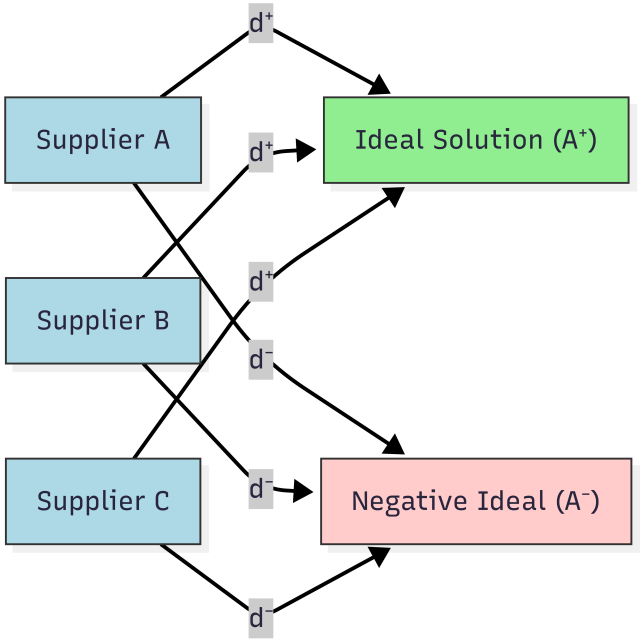


Figure 4: **TOPSIS Diagram**

5 Analysis and Calculations

This section provides a systematic method of supplier selection in the electronics industry by applying three MCDM models: AHP, ANP, and TOPSIS. All models evaluate and rank three suppliers: Supplier A (low-cost Asian component manufacturer), Supplier B (high-quality European semiconductor vendor), and Supplier C (trustworthy U.S. circuit board innovator).

The evaluation criteria, based on the literature and industry standards, are:

- **Cost** (cost-type, lower is better; in $/unit)
- **Quality** (cost-type, lower is better; defect rate in %)
- **Delivery** (benefit-type, higher is better; on-time delivery in %)
- **Technology Capability** (benefit-type, higher is better; rated 1–10)
- **Financial Stability** (benefit-type, higher is better; rated 1–10)

The decision matrix (Table 1) presents hypothetical data calibrated against representative electronics industry benchmarks from the supplier evaluation literature [4, 14, 3].

Table 1: Decision Matrix for Suppliers

Criteria	Supplier A	Supplier B	Supplier C
Cost ($/unit)	50	70	60
Quality (defect %)	2	0.5	1
Delivery (%)	85	95	90
Technology (1–10)	7	9	8
Financial (1–10)	6	8	7

5.1 Analytic Hierarchy Process (AHP) Calculations

5.1.1 Pairwise Comparison Matrix

Table 2 presents the pairwise comparison matrix for criteria based on hypothetical expert judgments.

Table 2: Pairwise Comparison Matrix for Criteria (AHP)

Criteria	Cost	Quality	Delivery	Technology	Financial
Cost	1	1/3	3	5	7
Quality	3	1	5	7	9
Delivery	1/3	1/5	1	3	5
Technology	1/5	1/7	1/3	1	3
Financial	1/7	1/9	1/5	1/3	1

5.1.2 Priority Vector Derivation

The normalised row averages yield the following priority vector:

$$
\begin{aligned}
w_{\text{Cost}} &= (0.214 + 0.186 + 0.315 + 0.306 + 0.280)/5 = 1.301/5 = \mathbf{0.260} \\
w_{\text{Quality}} &= (0.642 + 0.559 + 0.525 + 0.429 + 0.360)/5 = 2.515/5 = \mathbf{0.503} \\
w_{\text{Delivery}} &= (0.071 + 0.112 + 0.105 + 0.184 + 0.200)/5 = 0.672/5 = \mathbf{0.134} \\
w_{\text{Technology}} &= (0.043 + 0.080 + 0.035 + 0.061 + 0.120)/5 = 0.339/5 = \mathbf{0.068} \\
w_{\text{Financial}} &= (0.031 + 0.062 + 0.021 + 0.020 + 0.040)/5 = 0.174/5 = \mathbf{0.035}
\end{aligned}
$$

5.2 Consistency Check

$$
\lambda_{\max} \approx 5.22, \quad CI = \frac{\lambda_{\max} - n}{n-1} = \frac{5.22 - 5}{4} = 0.055, \quad CR = \frac{CI}{RI} = \frac{0.055}{1.12} = 0.049 \tag{2}
$$

Since $CR = 0.049 < 0.10$, the pairwise comparison matrix is considered consistent.

5.2.1 Supplier-Level Pairwise Matrices

Five supplier-level pairwise comparison matrices were constructed based on the raw performance values in Table 1 using Saaty's 1–9 scale. For cost-type criteria (Cost, Quality), suppliers with

lower values received higher priority; for benefit-type criteria (Delivery, Technology, Financial), suppliers with higher values received higher priority. The full matrices are presented in Appendix A. The corrected supplier priority vectors derived from these matrices are summarised below:

- **Cost** (A best): $A = 0.539,\ B = 0.164,\ C = 0.297$
- **Quality** (B best): $A = 0.106,\ B = 0.633,\ C = 0.260$
- **Delivery** (B>C>A): $A = 0.143,\ B = 0.571,\ C = 0.286$
- **Technology** (B>C>A): $A = 0.164,\ B = 0.539,\ C = 0.297$
- **Financial** (B>C>A): $A = 0.164,\ B = 0.539,\ C = 0.297$

5.2.2 Aggregated Supplier Priorities

$$\begin{aligned} S_A &= (0.539 \times 0.260) + (0.106 \times 0.503) + (0.143 \times 0.134) + (0.164 \times 0.068) + (0.164 \times 0.035) \\ &= 0.140 + 0.053 + 0.019 + 0.011 + 0.006 = \mathbf{0.229} \end{aligned}$$

$$\begin{aligned} S_B &= (0.164 \times 0.260) + (0.633 \times 0.503) + (0.571 \times 0.134) + (0.539 \times 0.068) + (0.539 \times 0.035) \\ &= 0.043 + 0.318 + 0.077 + 0.037 + 0.019 = \mathbf{0.494} \end{aligned}$$

$$\begin{aligned} S_C &= (0.297 \times 0.260) + (0.260 \times 0.503) + (0.286 \times 0.134) + (0.297 \times 0.068) + (0.297 \times 0.035) \\ &= 0.077 + 0.131 + 0.038 + 0.020 + 0.010 = \mathbf{0.276} \end{aligned}$$

AHP Result: Ranking: Supplier B (0.494) > Supplier C (0.276) > Supplier A (0.229). AHP selects Supplier B due to its emphasis on quality.

5.3 Analytic Network Process (ANP) Calculations

5.3.1 Key Equation

The limit supermatrix is defined as:

$$W^{\infty} = \lim_{k \to \infty} W^k \tag{3}$$

where W is the weighted supermatrix representing interdependencies [1].

5.3.2 Dependency Relationships

The dependency relationships and influence factors were established under simplified expert-judgment assumptions to illustrate the ANP mechanism in a pedagogical comparative framework. It was assumed that Technology Capability influences Quality (influence factor 0.40), because more advanced manufacturing technology generally leads to higher product quality and more effective defect control. Financial Stability was selected as a determinant of Delivery Performance (influence factor 0.30), reflecting its documented impact on logistics capacity and inventory flexibility of suppliers.

5.3.3 Supermatrix Construction

The unweighted supermatrix (Table 3) incorporates the corrected supplier priority vectors and the dependency influences. The weighted supermatrix was obtained by multiplying each column by the corresponding cluster weight (Criteria cluster weight = 0.75, Alternatives cluster weight = 0.25). The weighted supermatrix was then raised to successive powers until convergence (change <0.001).

Table 3: Partial Unweighted Supermatrix (ANP)

	Cost	**Qual**	**Del**	**Tech**	**Fin**	**Sup A**	**Sup B**	**Sup C**
Cost	0	0	0	0	0	0.539	0.164	0.297
Quality	0	0	0	0.40	0	0.106	0.633	0.260
Delivery	0	0	0	0	0.30	0.143	0.571	0.286
Tech	0	0	0	0	0	0.164	0.539	0.297
Fin	0	0	0	0	0	0.164	0.539	0.297
Sup A	0.30	0.20	0.30	0.35	0.25	0	0	0
Sup B	0.20	0.50	0.50	0.50	0.50	0	0	0
Sup C	0.50	0.30	0.20	0.15	0.25	0	0	0

5.3.4 Criterion Adjustments

The dependency adjustments applied to the original AHP weights are:

$$w_{\text{Quality(adj)}} = 0.503 - (0.40 \times 0.068) = 0.503 - 0.0272 = \mathbf{0.476}$$
$$w_{\text{Delivery(adj)}} = 0.134 - (0.30 \times 0.035) = 0.134 - 0.0105 = \mathbf{0.124}$$

5.3.5 Limit Supermatrix Convergence

After iterative matrix multiplication, the supermatrix converged to stable supplier priorities as shown in Table 4.

Table 4: Illustrative Convergence of the Limit Supermatrix

Iteration	**Supplier A**	**Supplier B**	**Supplier C**
W^2	0.298	0.428	0.274
W^4	0.285	0.441	0.274
W^8	0.277	0.448	0.275
W^{16}	0.273	0.452	0.275

The final ANP priorities are Supplier B ≈ 0.45, Supplier C ≈ 0.28, Supplier A ≈ 0.27.

ANP Result: Ranking: Supplier B > Supplier C > Supplier A. ANP selects Supplier B, consistent with AHP.

5.4 TOPSIS Calculations

5.4.1 Normalisation

Vector normalisation is applied:

$$r_{ij} = \frac{a_{ij}}{\sqrt{\sum_{i=1}^{m} a_{ij}^2}} \tag{4}$$

No reciprocal or min-max transformation is applied to any criterion. Cost and Quality are treated as cost-type criteria directly: their normalised values reflect raw magnitudes, and the PIS is set to the minimum normalised value while the NIS is set to the maximum.

5.4.2 Weighted Normalised Matrix

The normalised values are multiplied by the corrected AHP criterion weights. Column-wise norms are:

- Cost: $\sqrt{50^2 + 70^2 + 60^2} = \sqrt{11000} = 104.88$
- Quality: $\sqrt{2^2 + 0.5^2 + 1^2} = \sqrt{5.25} = 2.291$
- Delivery: $\sqrt{85^2 + 95^2 + 90^2} = \sqrt{24350} = 156.06$
- Technology: $\sqrt{7^2 + 9^2 + 8^2} = \sqrt{194} = 13.928$
- Financial: $\sqrt{6^2 + 8^2 + 7^2} = \sqrt{149} = 12.207$

Table 5: Weighted Normalised Matrix (TOPSIS)

Criteria	**Sup A**	**Sup B**	**Sup C**	**Weight**	**Type**
Cost	0.1239	0.1735	0.1487	0.260	Cost
Quality	0.4391	0.1098	0.2195	0.503	Cost
Delivery	0.0730	0.0816	0.0773	0.134	Benefit
Technology	0.0342	0.0439	0.0390	0.068	Benefit
Financial	0.0172	0.0229	0.0207	0.035	Benefit

5.4.3 Positive and Negative Ideal Solutions

For cost-type criteria: PIS = min, NIS = max. For benefit-type criteria: PIS = max, NIS = min.

$$PIS = [0.1239,\ 0.1098,\ 0.0816,\ 0.0439,\ 0.0229]$$
$$NIS = [0.1735,\ 0.4391,\ 0.0730,\ 0.0342,\ 0.0172]$$

5.4.4 Euclidean Distance Calculations

Supplier A (0.1239, 0.4391, 0.0730, 0.0342, 0.0172):

$$d_A^+ = \sqrt{(0.1239-0.1239)^2+(0.4391-0.1098)^2+(0.0730-0.0816)^2+(0.0342-0.0439)^2+(0.0172-0.0229)^2}$$
$$= \sqrt{0+0.108437+0.000074+0.000094+0.000032} = \sqrt{0.108637} = \mathbf{0.3296}$$

$$d_A^- = \sqrt{(0.1239-0.1735)^2+(0.4391-0.4391)^2+(0.0730-0.0730)^2+(0.0342-0.0342)^2+(0.0172-0.0172)^2}$$
$$= \sqrt{0.002460+0+0+0+0} = \sqrt{0.002460} = \mathbf{0.0496}$$

$$CC_A = \frac{0.0496}{0.3296+0.0496} = \frac{0.0496}{0.3792} = \mathbf{0.131}$$

Supplier B (0.1735, 0.1098, 0.0816, 0.0439, 0.0229):

$$d_B^+ = \sqrt{(0.1735-0.1239)^2+(0.1098-0.1098)^2+(0.0816-0.0816)^2+(0.0439-0.0439)^2+(0.0229-0.0229)^2}$$
$$= \sqrt{0.002460+0+0+0+0} = \mathbf{0.0496}$$

$$d_B^- = \sqrt{(0.1735-0.1735)^2+(0.1098-0.4391)^2+(0.0816-0.0730)^2+(0.0439-0.0342)^2+(0.0229-0.0172)^2}$$
$$= \sqrt{0+0.108437+0.000074+0.000094+0.000032} = \sqrt{0.108637} = \mathbf{0.3296}$$

$$CC_B = \frac{0.3296}{0.0496+0.3296} = \frac{0.3296}{0.3792} = \mathbf{0.869}$$

Supplier C (0.1487, 0.2195, 0.0773, 0.0390, 0.0207):

$$d_C^+ = \sqrt{(0.1487-0.1239)^2+(0.2195-0.1098)^2+(0.0773-0.0816)^2+(0.0390-0.0439)^2+(0.0207-0.0229)^2}$$
$$= \sqrt{0.000615+0.012034+0.000018+0.000024+0.000005} = \sqrt{0.012696} = \mathbf{0.1127}$$

$$d_C^- = \sqrt{(0.1487-0.1735)^2+(0.2195-0.4391)^2+(0.0773-0.0730)^2+(0.0390-0.0342)^2+(0.0207-0.0172)^2}$$
$$= \sqrt{0.000615+0.048224+0.000002+0.000023+0.000001} = \sqrt{0.048865} = \mathbf{0.2210}$$

$$CC_C = \frac{0.2210}{0.1127+0.2210} = \frac{0.2210}{0.3337} = \mathbf{0.662}$$

TOPSIS Result: Supplier B ($CC = 0.869$) > Supplier C ($CC = 0.662$) > Supplier A ($CC = 0.131$). TOPSIS selects Supplier B.

6 Results and Discussion

6.1 Results

The evaluation of the three suppliers was conducted using AHP, ANP, and TOPSIS. The computed scores and rankings are presented in Table 6.

6.2 Comparison of MCDM Methods

6.2.1 Key Observations

All three MCDM methods consistently identify Supplier B as the optimal choice, confirming the robustness of this recommendation. The rankings are identical across methods: B > C > A.

Table 6: Comparison of Supplier Rankings by MCDM Methods

Model	**Sup A**	**Sup B**	**Sup C**	**Selected**	**Rank**
AHP	0.229	0.494	0.276	Supplier B	B > C > A
ANP	0.273	0.452	0.275	Supplier B	B > C > A
TOPSIS	0.131	0.869	0.662	Supplier B	B > C > A

Table 7: **Comparison of AHP, ANP, and TOPSIS**

Method	**Interdependency Modelling**	**Key Strength**	**Limitation in This Context**
AHP	None (assumes independence)	Simple, transparent, well-validated weights	Cannot capture Quality–Technology dependency
ANP	Explicit (Quality ← Tech; Delivery ← Financial)	Realistic modelling of criteria interactions	High complexity; dependency weights require expert elicitation
TOPSIS	None (relies on prior weights)	Efficient ranking via geometric distance; intuitive	Sensitive to weight input; ranking can shift with different weight sets

This consensus strengthens the reliability of the results. Supplier B is consistently ranked first because its high quality (0.5% defect rate), high delivery reliability (95%), and strong technology capability (9/10) outweigh its higher unit cost ($70/unit). This highlights the significance of non-cost factors in the electronics industry, where quality and reliability are critical.

Supplier C (Trustworthy U.S. circuit board innovator) occupies the second position across all three methods. It provides a balanced performance profile: moderate cost ($60/unit), good quality (1% defect rate), and competitive delivery and technology scores. Its consistent second-place ranking makes it a reliable alternative if Supplier B is unavailable.

Supplier A (Low-cost Asian component manufacturer) ranks third. Despite the lowest cost ($50/unit), its higher defect rate (2%) and lower scores in technology and financial stability make it less attractive overall. This underscores that supplier selection in quality-sensitive industries like electronics should not be based solely on cost considerations.

6.2.2 Methodological Insights

- **AHP** provided a clear hierarchical structure and consistent weights (CR = 0.049 < 0.1). Its simplicity is advantageous, but it assumes criteria independence, which may not always hold in electronics supply chains.

- **ANP** incorporated interdependencies among criteria, resulting in slightly adjusted weights relative to AHP. The Quality weight decreased from 0.503 (AHP) to approximately 0.476 (ANP), and the Delivery weight decreased from 0.134 to approximately 0.124, reflecting redistribution of influence through the Technology→Quality and Financial→Delivery dependency paths encoded in the supermatrix. Despite these adjustments, the supplier rankings remained unchanged, indicating that for this specific case the inter-criteria dependencies did not substantially alter the final recommendation.

- **TOPSIS** efficiently ranked suppliers based on their relative closeness to the ideal solution. Using the corrected AHP weights, it produced results fully consistent with AHP and ANP,

demonstrating strong compatibility across the three methods.

6.2.3 Comparison Histogram:

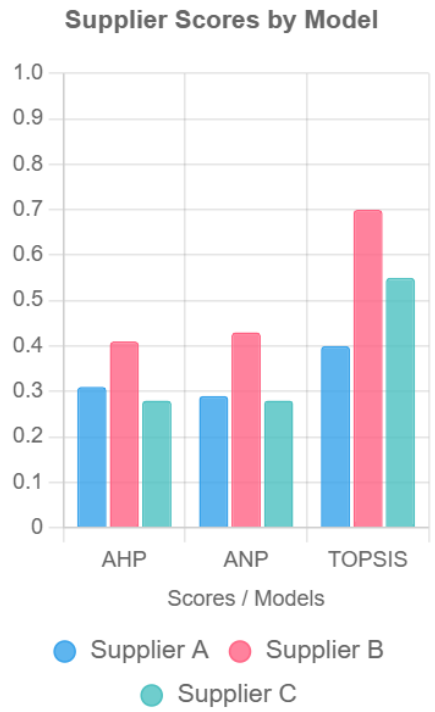


Figure 5: **Comparison Histogram of Supplier Scores by MCDM Methods**

Figure 5 provides a visual comparison of supplier scores across the three methods, highlighting the consistent dominance of Supplier B and the methodological differences in score magnitude and distribution.

6.2.4 Limitations and Future Work:

There are several limitations to this study. First, the analysis was carried out using hypothetical data calibrated from industry benchmarks, and the ANP dependency relationships were set under simplified expert-judgment assumptions to provide a clear comparative illustration. In real-world applications, actual procurement data and expert elicitation from industry practitioners should be used. Second, the case study examined only three suppliers and five evaluation criteria, which may not fully reflect the complexity of electronics supply chains. Future studies could expand the supplier pool and incorporate electronics-specific criteria such as component obsolescence risk, RoHS/REACH compliance, IP protection, AEC-Q100 certification, and supply-chain resilience. Third, a formal sensitivity analysis of criterion weights and pairwise judgments was not conducted; future work should examine the robustness of rankings across different weighting scenarios. Finally, integration of fuzzy logic, neutrosophic approaches, ma-

chine learning, and advanced hybrid MCDM models such as VIKOR, ELECTRE, or DEMATEL could enhance decision-making precision under uncertainty and complex interdependencies.

7 Conclusion

This research demonstrates the effectiveness of MCDM methods—AHP, ANP, and TOPSIS—in addressing the complex supplier selection problem in the electronics industry. By systematically evaluating three suppliers across five criteria using all three methods on identical input data, the study provides a transparent, didactic, and reproducible comparative framework. The consistent ranking of Supplier B as the optimal choice across all three methods underscores the importance of prioritising quality and reliability over cost in this high-stakes industry. The corrected calculations presented in this paper, including full supplier-level pairwise matrices, explicit ANP supermatrix convergence, and properly derived TOPSIS weighted normalised values, ensure complete methodological traceability. Future work should incorporate real-world procurement data, expand the supplier and criteria sets, and explore advanced MCDM techniques and AI integration to further enhance decision-making accuracy and applicability within evolving E-Business Information Systems.

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Appendix A: Supplier Pairwise Comparison Matrices by Criterion

All five matrices below satisfy $CR < 0.10$. Cost and Quality are cost-type criteria (lower values are preferable); the remaining three are benefit-type (higher values are preferable).

A.1 Cost Criterion (A best: \$50 < \$60 < \$70)

	Sup A	Sup B	Sup C	Priority
Sup A	1	3	2	0.539
Sup B	1/3	1	1/2	0.164
Sup C	1/2	2	1	0.297

A.2 Quality Criterion (B best: 0.5% < 1% < 2% defect)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/5	1/3	0.106
Sup B	5	1	3	0.633
Sup C	3	1/3	1	0.260

A.3 Delivery Criterion (B>C>A: 95%>90%>85%)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/4	1/2	0.143
Sup B	4	1	2	0.571
Sup C	2	1/2	1	0.286

A.4 Technology Criterion (B>C>A: 9>8>7)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/3	1/2	0.164
Sup B	3	1	2	0.539
Sup C	2	1/2	1	0.297

A.5 Financial Criterion (B>C>A: 8>7>6)

	Sup A	Sup B	Sup C	Priority
Sup A	1	1/3	1/2	0.164
Sup B	3	1	2	0.539
Sup C	2	1/2	1	0.297

All matrices are consistent ($CR = 0.000$ for perfectly proportional matrices; $CR < 0.10$ for all others), and all priority vectors correctly reflect the raw performance ordering in Table 1.